



Step-by-Step Firebase Integration

1. Set Up Firebase

1. Go to [Firebase Console](#)
 2. Create a new project.
 3. Go to **Realtime Database** > **Create Database** → Start in test mode.
 4. Go to **Project Settings** > **Service Accounts** > **Generate new private key** → Download the .json file.
-

2. Install Firebase Admin SDK

```
pip3 install firebase-admin
```

3. Updated Python Code with Firebase Integration

```
import time
import Adafruit_DHT
import RPi.GPIO as GPIO
import firebase_admin
from firebase_admin import credentials, db

# ===== FIREBASE SETUP =====
cred = credentials.Certificate('/home/pi/firebase-adminsdk.json') # <--
update path to your .json file
firebase_admin.initialize_app(cred, {
    'databaseURL': 'https://your-database.firebaseio.com/' # <-- update
with your Firebase DB URL
})

# Firebase DB reference
ref = db.reference('/air_quality_log')

# ===== GPIO & Sensor Setup =====
GPIO.setmode(GPIO.BCM)
MQ135_PIN = 17
GPIO.setup(MQ135_PIN, GPIO.IN)

DHT_SENSOR = Adafruit_DHT.DHT22
DHT_PIN = 4

# ===== Main Loop =====
while True:
    gas_status = GPIO.input(MQ135_PIN) # 0 = poor air, 1 = good air
    humidity, temperature = Adafruit_DHT.read_retry(DHT_SENSOR, DHT_PIN)

    data = {
        'air_quality': 'Good' if gas_status else 'Poor',
        'temperature_c': round(temperature, 1) if temperature is not None
    else None,
```

```
        'humidity': round(humidity, 1) if humidity is not None else None,
        'timestamp': time.strftime("%Y-%m-%d %H:%M:%S")
    }

    print(f"Uploading: {data}")
    ref.push(data)

    time.sleep(10)
```

Sample Firebase Data Structure

```
air_quality_log: {
  "-NabcXYZ123": {
    "air_quality": "Poor",
    "temperature_c": 30.5,
    "humidity": 65.2,
    "timestamp": "2025-05-02 14:25:10"
  }
}
```

Make sure to:

- Replace '/home/pi/firebase-adminsdk.json' with the correct path to your Firebase credentials file.
 - Replace 'https://your-database.firebaseio.com/' with your actual **database URL**.
-